

From Material Turnover to Causal Memory: A Prospective Falsification Ladder for Organizational Continuity in a Simulated Droplet

[AUTHOR NAME REQUIRED]¹

¹[AFFILIATION REQUIRED] ORCID: [ORCID IF AVAILABLE]

July 2026 (Revised manuscript, V2, incorporating three referee reports)

Abstract

Biological individuals persist while their constituent material is continually replaced, which motivates the idea that *organization*, rather than matter, is a substrate of biological memory and identity. We ask a deflationary, falsifiable version of this question in a simulated reaction-transport droplet endowed with a designed two-component intensive memory field $m = (m_1, m_2)$ coupled to nutrient uptake ($m_+ = m_1 + m_2$) and attractant production ($m_- = m_1 - m_2$). Over a preregistered ladder of interventions evaluated prospectively, (and, we stress at the outset, without any claim of individual identity), with grouped leave-history-out statistics, sealed hold-out families, kill-switches, and continuity-safe entity tracking, we establish one positive result and a matched negative boundary. **Positive:** under a globally imposed nutrient history, many persistent droplets locally encode a similar engineered cumulative-experience coordinate h_1 ; a *continuously tracked* droplet retains this internal causal state through substantial constituent turnover (longitudinal hold-out on untouched seeds: h_1 held-out $R^2 = 0.98$, 95% CI $[0.97, 1.00]$ at old-material fraction $M \approx 0.19$, with 36/36 track survival), and h_1 is causally expressed through uptake and transplantable into a fresh standardized body ($R^2 = 0.61$), exceeding body-size/mass baselines after turnover. Crucially h_1 is *global* in informational content: it is decodable from essentially any large droplet, and therefore does **not** individuate droplets. **Negative:** a second initially decodable coordinate h_2 (temporal order / value dispersion) fails qualification — its causal response is $\approx$ one-dimensional, and its deep-turnover held-out $R^2 = 0.34$ (< 0.50) under continuity-safe tracking. Our methodological contribution is a falsification ladder that separates decodability, storage, causal expression, material-turnover survival, longitudinal continuity, and individuation, together with a transparent correction chain (grouped-evaluation leakage; seed-family instability; statistical power; largest-component reselection versus continuity-safe tracking). We claim no life, agency, identity, individuation, reproduction, or heredity.

1 Introduction

An organism replaces most of its molecules many times over its life, yet remains, in a functional sense, the same individual. This everyday fact underlies a family of ideas — from cybernetics and autopoiesis to contemporary theories of biological individuality — in which persistence of *organization* rather than of *matter* is treated as a candidate substrate for biological memory and identity [1–3]. If an organized, self-maintaining entity could record aspects of its history in a way that outlives its constituent material, that record would be a natural minimal model of organizational

memory. Our goal in this paper is to render this intuition precise, testable, and — where the evidence demands — falsifiable, in a tractable simulation.

We adopt three deliberately stringent criteria for what should count as a memory of experience, as opposed to a mere statistical shadow of it. First, a history variable being *decodable* from a system is not sufficient: a decoder can exploit residual environment, current body size, laboratory-frame position, or simply the fact that a globally imposed history is written similarly into every entity in the world. Decodability is necessary but weak. Second, a memory worth the name must be *causal*: it must shape the system’s dynamics, in the interventionist sense that ablating or transplanting it changes downstream physical behaviour [4–6]. Third, if the memory is to be *organizational* rather than merely material, it must survive substantial replacement of the constituent material of the entity that carries it. To these we add a fourth, sharper property that is often conflated with the third: *individuation*, i.e. whether the memory makes one entity distinguishable from another by its particular history. Individuation requires strictly more than persistence, and we do not claim it here.

These criteria motivate an experimental programme rather than a single measurement. Decodability, causal expression, and turnover survival can each hold or fail independently, and an honest evaluation must test them separately and prospectively. We therefore construct a “falsification ladder”: a sequence of preregistered experiments, each with a sealed prospective family and an explicit pass/fail gate, that climbs from the weakest property (decodability at rest) toward the strongest (a causal, transplantable, turnover-surviving, history-individuating coordinate). At each rung we attempt to break the positive reading with the strongest available trivial explanation before accepting it, and we retain and report negative results as first-class outcomes.

The system we study is a simulated “scaffold” droplet: a periodic two-dimensional reaction-transport medium in which a localized, viable, self-maintaining density field advects toward a self-produced attractant, grows on a nutrient, and dies. On top of a frozen version of this substrate we engineer a bounded, distributed memory field that is written by a local experience signal, maintained on two timescales, inherited by newly grown material, and read back through two pre-existing physical channels (uptake and attractant production). We stress from the outset that this memory is *engineered*, not emergent: our contribution is not a new memory mechanism but a disciplined test of the *causal and material status* of a history coordinate carried by such a mechanism under turnover, and a demonstration of how several plausible positive readings fail.

Two results anchor the paper. The first is positive but carefully scoped: a one-dimensional cumulative-experience coordinate, which we call h_1 , is prospectively decodable, causally expressed through uptake, transplantable into a fresh standardized body, and — when a single droplet is followed with a continuity-safe tracker — retained through the replacement of roughly four-fifths of its material. However, because the driving history is imposed globally, h_1 is decodable from essentially any large droplet in the world; it is a memory of the world’s history carried locally, not a marker of individual identity. The second result is a matched negative boundary: a second coordinate h_2 , encoding temporal order, is decodable at rest but fails to qualify as an independent, causal, turnover-surviving dimension. The interplay of these two results, and the corrections required to reach them, constitute the paper’s contribution.

A methodological theme runs throughout. Reaching the scoped positive and the clean negative required correcting several seductive errors: a replicate-leaking cross-validation that inflated an apparent high-dimensional result [7]; a single seed family whose behaviour did not generalize; an underpowered deep-turnover estimate; and — exposed by a collaborator’s visual inspection of an animation — a largest-component *reselection* tracker that manufactured spurious entity continuity. We treat these corrections not as embarrassments but as the empirical substance of a falsification ladder, and we devote explicit methods and results subsections to the tracking correction because

it changed the scientific record.

To make the programme concrete, we frame six competing readings that any candidate coordinate must survive, and we attach an experiment to each. A coordinate could be (i) a genuine causal, transplantable, turnover-surviving, individuating memory; or it could be a mere (ii) material residue carried by the original constituents, (iii) environmental residue read from persistent nutrient or attractant fields, (iv) laboratory-frame artefact exploiting absolute position or orientation, (v) measurement artefact visible only through an inadequate readout, or (vi) a stored but causally silent pattern. These readings are not rhetorical: each makes a distinct, falsifiable prediction under a specific control (turnover, environmental reset, translation/rotation, alternative readout, targeted ablation), and much of the paper consists of executing those controls. The discipline of naming the trivial explanations *before* testing, and of trying hardest to confirm them, is what separates a falsification ladder from a search for a positive result.

Throughout, we distinguish sharply between what is *observed* (a decode value, a confidence interval, an ablation contrast), what is *inferred* (a mechanistic reading consistent with the observations), and what remains *speculative* (an interpretation not yet tested). We also distinguish the *writing-map* capacity (the rank of the map from history to stored memory), the *memory-state* dimensionality (the rank of the stored field), the *observability* of a coordinate (whether a decoder can read it at a chosen window), and the *causal-response* dimensionality (how many independent axes the memory drives in the physics). Conflating these four is the single most common way to over-read a positive decode, and keeping them separate is a precondition for the corrections we report. We work in simulation deliberately. A wet-lab protocell offers physical realism but poor experimental control: one cannot erase a droplet’s memory exactly, transplant an internal state into a standardized body, ablate a single readout channel while holding everything else fixed, or replay a trajectory deterministically. These operations — which distinguish a causal memory from a correlate — are trivial in a frozen simulator and are why a simulation can answer the causal and material questions more sharply than a physical system can, at the cost of realism. The trade is explicit: we buy interventional and statistical rigor and pay with simulator specificity, and we return to the substrate-versus-architecture question in the discussion.

2 Related work

Our substrate belongs to the long tradition of reaction-diffusion and dissipative pattern formation initiated by Turing [8] and developed through activator-inhibitor theory [9], positional information [10], and the modern synthesis of nonequilibrium pattern formation [11–13]. The local excitable chemistry we use is of FitzHugh-Nagumo type [14, 15]. Unlike this literature, our interest is not the patterns themselves but a superimposed, bounded memory field and its causal and material status.

The memory we engineer is an attractor-like internal state. Distributed attractor memories are classical in theoretical neuroscience [16], and the dynamical-systems notions of attractors and observability frame what it means for a hidden state to be recoverable and consequential [17, 18]. We borrow the vocabulary of observability but reach a deflationary conclusion: recoverability of a hidden coordinate is not, by itself, evidence of individual memory.

At the biological end, our droplet is a caricature of an active droplet or protocell — an idealized model *inspired by*, not quantitatively corresponding to, such systems. Chemically active droplets can grow and divide in ways reminiscent of cell proliferation [19], liquid-liquid phase separation organizes intracellular biochemistry into condensates [20–22], and minimal protocells can encapsulate, grow, and divide [23, 24]. This literature largely concerns *physical* self-organization and reproduc-

tion; it does not typically ask whether such droplets carry a *causal, turnover-surviving memory of their individual experience*, which is our question. Molecular biology does provide concrete memory mechanisms — bistable genetic and chemical switches with hysteresis [25–28] — and we use these as a conceptual benchmark for what a qualified second, independent memory dimension would look like, and as motivation for why our (failed) h_2 would have needed bistable, protected maintenance rather than passive linear averaging to persist.

Conceptually, the paper engages theories of biological individuality and organizational identity [1–3, 29, 30] and the artificial-life programme’s open problems around the emergence and persistence of individuals [31]. Our stance is empirical and deflationary: rather than argue for a definition of individuality, we operationalize necessary conditions (causality, transplantability, turnover survival, individuation) and test them. Methodologically we lean on causal inference [4–6], the bootstrap for grouped uncertainty [32], preregistration and falsification [33, 34], the theory of leakage in predictive evaluation [7], and classical connected-component and particle-tracking methods [35, 36], the last of which is central to the tracking correction we report.

Two contrasts with prior modelling deserve emphasis. First, in the active-droplet and protocell literature the object of interest is typically the *physical* life cycle — growth, shape instability, division [19, 24] — and memory, when discussed, is chemical composition rather than a decodable, causally-tested history coordinate. We instead hold the physics frozen and ask a measurement question about an engineered internal coordinate, which lets us apply the machinery of prospective evaluation and causal intervention. Second, in synthetic biology, durable memory is achieved by *bistable* circuits whose hysteresis protects a state across perturbation and division [25–27]. Our maintenance terms are, by contrast, linear and averaging; the comparison predicts — correctly, as it turns out — that a dispersion-coded second coordinate should not survive turnover, and it points to bistable maintenance as the missing ingredient. The observability lens clarifies why the two coordinates behave so differently under turnover [18]. The cumulative coordinate h_1 is encoded in the *level* of the memory, a quantity that growth and uniform death preserve and that the smoothing terms do not erase; it is therefore robustly observable through turnover. The order coordinate h_2 is encoded in the *dispersion* of the memory, a quantity that linear low-pass maintenance and fresh, history-independent growth actively degrade; it is therefore observable at rest but not through deep turnover. This is not a subtle statistical accident but a structural property of the maintenance operator, and it is exactly the property that a bistable, hysteretic rule — which protects a discrete distinction rather than averaging a continuous one — would change [25, 27, 28]. In this sense the negative result is not merely a null; it is interpretable against an established mechanistic benchmark. The conceptual target — persistence of an individual through material replacement — is old, from the ship of Theseus to the observation that most somatic constituents turn over on timescales far shorter than an organism’s life. Modern philosophy of biology distinguishes physiological individuality (a cohesive self-maintaining metabolic unit) from evolutionary individuality (a unit of selection with reproduction and heredity) [3], and organizational accounts locate identity in a self-producing organization rather than its matter [2, 29]. Our droplet speaks only to a minimal physiological notion: whether a self-maintaining unit can carry a causal record of its experience across turnover. It says nothing about evolutionary individuality, which requires reproduction and heredity that we neither implement nor claim; a turnover-surviving causal state is sometimes over-read as evidence of the evolutionary notion, and our results support only the weaker reading, for a globally shared history. We are explicit about novelty. The mechanism (m_+/m_- readouts, two-timescale integration) is engineered and modest; we do not present it as a new memory principle. The contribution is the disciplined *evaluation*: a prospective, kill-switched falsification ladder that repeatedly resists the temptation to over-read a positive decode, together with a transparent chain of corrections — replicate leakage, seed-family instability, statistical power, and largest-component

reselection — each of which is a recurring hazard in computational studies of simulated organisms. A reader who takes nothing else from the paper should take the methodological pattern: intervene, group, seal, power, track, and preregister a stopping rule, so that a marginal signal cannot be inflated into a claim of individual memory or life.

3 Model

The substrate is a periodic 64×64 lattice evolved with timestep $dt = 0.1$ (Table 1, Table 2). It carries a coarse density field $\rho(x, t)$, an internal excitable chemistry (U, V) , a diffusing nutrient N , and a self-produced attractant c . Density is transported advectively up attractant gradients, grows where nutrient is available and crowding is low, and decays; the internal (U, V) chemistry follows a FitzHugh–Nagumo-type excitable law [14, 15]. This “scaffold” is frozen throughout the paper (its implementation hash is recorded in the reproducibility appendix); all experiments only add or read a memory field on top of it, or intervene on that memory.

Memory writing. The engineered memory is a bounded intensive field $m = (m_1, m_2)$, stored extensively as $M_f = \rho m$ so that it is transported with the density exactly as the internal chemistry is. It is written by a local experience signal

$$\Psi = \tanh(k_{\text{exp}}(N - c) + k_{\text{up}}(\text{uptake} - \overline{\text{uptake}})), \quad (1)$$

which compares local nutrient to attractant and local uptake to its spatial mean, so that Ψ encodes a bounded “how nutritive and how surprising is it here” signal. Each memory component is a leaky integrator of Ψ with its own forgetting timescale, together with local templating toward a neighbourhood consensus and weak diffusion:

$$\dot{m}_k = \eta_w \Psi - \eta_{d,k} m_k + \eta_t(\langle m_k \rangle_{\text{nbr}} - m_k) + D_m \nabla^2 m_k, \quad m_k \in [-1, 1], \quad k \in \{1, 2\}. \quad (2)$$

The two components differ only in their forgetting rate ($\eta_{d,1} \gg \eta_{d,2}$; a fast and a slow integrator). Newly grown material inherits the local intensive memory, $M_f \mapsto g m$ (with g the local growth), and death scales all fields uniformly; consequently growth and death approximately *preserve* the intensive memory, a fact that matters for the turnover analysis.

Memory readout. The memory is read through two pre-existing physical channels via the sum and difference coordinates $m_+ = m_1 + m_2$ and $m_- = m_1 - m_2$:

$$\text{uptake} \rightarrow \text{uptake} (1 + \lambda_+ \tanh m_+), \quad (3)$$

$$\text{attractant production} \propto 1 + \lambda_- \tanh m_-. \quad (4)$$

With $\lambda_- = 0$ the model reduces bit-identically to a single-channel predecessor, and with $\lambda_+ = \lambda_- = 0$ it reduces to the frozen scaffold; these exact reductions are used as backward-compatibility tests. The two readouts are the only causal pathways from memory to physics.

History coordinates. Experience is imposed as a two-phase *global* nutrient history: during an early phase the whole field receives $N \mapsto a_{\text{early}}$ per step for T steps, and during a late phase $N \mapsto a_{\text{late}}$ for T steps. From the two phase amplitudes we define two orthogonal history coordinates,

$$h_1 = a_{\text{early}} + a_{\text{late}} \quad (\text{cumulative dose}), \quad h_2 = a_{\text{late}} - a_{\text{early}} \quad (\text{temporal order}), \quad (5)$$

Table 1: State variables of the frozen scaffold and the engineered memory.

symbol	type	interpretation
ρ	field	coarse density of the droplet material
U, V	fields	internal excitable chemistry (FitzHugh–Nagumo type)
N	field	diffusing nutrient (globally driven by the history)
c	field	self-produced diffusing attractant
$m = (m_1, m_2)$	field	engineered intensive memory (fast / slow)
$M_f = \rho m$	field	extensive memory, transported with ρ
cohort C	field	passive pulse–chase tracer (never enters dynamics)

Table 2: Frozen C1c parameters (the selected writing configuration; unchanged in all reported analyses).

symbol	value	meaning
η_w	0.015	experience write gain
$\eta_{d,1}, \eta_{d,2}$	0.35, 0.006	fast / slow forgetting rates
η_t	0.010	local templating (4-neighbour consensus)
D_m	0.010	memory diffusion coefficient
$k_{\text{exp}}, k_{\text{up}}$	1.0, 1.0	experience–signal gains in Eq. (1)
λ_+, λ_-	0.25, 0.15	uptake / attractant coupling in Eqs. (3)–(4)
grid, dt	$64 \times 64, 0.1$	lattice, timestep

sampled independently over a viable amplitude band (Table 4). Because the drive is global, every droplet in the world experiences the same (h_1, h_2) ; the decoding task across a family therefore tests whether the *world’s* history is written into local memory, not whether droplets carry *distinct* histories.

Scaffold dynamics and boundary conditions. The frozen scaffold combines four ingredients. Transport is advective toward the self-produced attractant, so density accumulates where c is high, producing localized, motile droplets rather than a spread film. Growth is nutrient-limited and crowding-limited, $g \propto \rho N \max(0, 1 - \rho/\rho_{\text{max}})$, so entities saturate in size; death is a uniform decay. The internal (U, V) chemistry is excitable [14, 15], giving each droplet a bounded internal state that co-moves with the density. All fields are periodic in both directions, which is important for the tracking analysis: a droplet may straddle a boundary and appear at both edges while remaining a single connected component under wrap-aware labelling [35]. Because the memory is stored extensively as $M_f = \rho m$ and advected, diffused, and inherited exactly as U, V are, it is a bona fide internal field of the droplet rather than an externally imposed label; in particular no entity identity, cohort index, seed number, or history label ever enters Eqs. (1)–(4). This “no-leakage” property is enforced by construction and checked in every experiment. Two additional integrity tests bracket every result. Backward-compatibility: setting $\lambda_- = 0$ reproduces the single-channel predecessor bit-identically, and setting $\lambda_+ = \lambda_- = 0$ reproduces the frozen scaffold, so any measured effect is attributable to the memory coupling and not to an incidental change in the base dynamics. No-leakage: because history labels, cohort indices, seeds, and entity identities never enter Eqs. (1)–(4), a positive decode cannot be an artefact of a smuggled identity variable; the pulse–chase tracer that quantifies turnover is likewise inert, re-binning a passive cohort field without touching ρ, U, V, N, c .

4 Experimental methodology

Protocol. Each trial warms a body from a random seed for 2000 steps, erases the memory ($M_f = 0$), applies a two-phase history (Eq. 5; $T = 60$ per phase), settles for 20 steps, then relabels all extant material as “old” (cohort 0) and evolves the frozen dynamics forward with no further drive; material grown thereafter is “new” (cohort 1). The old-material fraction $M = \text{mass}_{\text{old}}/\text{mass}_{\text{total}}$ of the tracked entity quantifies turnover; the pulse-chase tracer is passive and never enters — “turnover” here is a bookkeeping construct on a passive cohort field, not a chemical exchange process, and the tracer ρ, U, V, N, c [36]. Interventions comprise erasure ($M_f = 0$), mean-field and structured transplant into a standardized erased recipient body, channel-specific readout ablation ($\lambda_+ \rightarrow 0$ or $\lambda_- \rightarrow 0$), and an exact-clone / stochastic ceiling (Table 3).

Entity detection and tracking. The substrate self-organizes into many small droplets. Connected components are labelled with a periodic (wrap-aware) algorithm and summarized by circular centroids (a *viable* entity is a single localized connected component above a minimum size threshold) [35]; all memory features used for decoding are permutation-invariant statistics of the memory values over an entity’s cells and are therefore translation- and reflection-invariant by construction, a property we verify directly (§5.1). The historical analyses selected, at each checkpoint, the *largest* connected component. As we show in §5.1, this is a per-frame *reselection* rule, not longitudinal tracking; we therefore additionally define a frozen longitudinal tracker that initializes once and thereafter follows the entity of maximum periodic mask overlap, censoring the track on loss and never using h_1, h_2, M , or any history label in its matching. All turnover claims are re-certified with this tracker.

Decoding and statistics. History coordinates are decoded from internal memory by ridge regression with grouped *leave-history-out* cross-validation: all replicate rows that share a history (across seeds) are held out together, so no replicate of a held-out history remains in training. This is essential because row-wise leave-one-out leaks through shared replicates and inflates apparent performance [7]. Development and prospective families are strictly separated; prospective families are generated and SHA-256-hashed before any selection and executed exactly once (Table 5); uncertainty is quantified by donor-level bootstrap 95% confidence intervals [32]; and every primary decode is compared against constant, shuffled-history, and same-law/different-seed nulls. Causal claims use interventions in the potential-outcomes and graphical sense [4–6]: an effect is causal if it changes under erasure, transplant, or channel ablation and collapses under a memory-inert control. Preregistration and explicit pass/fail kill-switches [33, 34] bound the programme so that mechanism development cannot continue indefinitely.

Decoder and score. For a history family we aggregate replicate rows to the history level and predict a coordinate $y \in \{h_1, h_2\}$ from a fixed feature vector ϕ (permutation-invariant memory statistics) by ridge regression, reporting the grouped leave-one-history-out coefficient of determination

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_{-i})^2}{\sum_i (y_i - \bar{y})^2}, \quad (6)$$

where $\hat{y}_{-i}$ is the prediction for history i from a model fit on all other histories. Preprocessing (centering, scaling, near-constant-column removal) is fit on the training fold only. Confidence intervals are obtained by resampling histories (donors) with replacement [32]. A coordinate is deemed decodable only if R^2 exceeds the constant, shuffled-history, and same-law/different-seed nulls, and, for certification, only if the lower confidence bound exceeds the preregistered threshold.

Longitudinal tracker. From the initial largest entity with cell set S_0 , the tracked entity at the next update is

$$S_{t+1} = \arg \max_{S \in \text{components}} \frac{|S \cap S_t|}{|S_t|}, \quad \text{lost if } \max_S \frac{|S \cap S_t|}{|S_t|} < \theta, \quad (7)$$

with overlap threshold $\theta = 0.1$; a tie within 10% marks the track *ambiguous* and censors it. The score in Eq. (7) depends only on geometry, never on h_1 , h_2 , M , or labels. Splits are handled by following the maximum-overlap fragment; merges keep the maximum-overlap component and are recorded as events. Longitudinal metrics are reported only for uncensored tracks, so a lost entity is never silently replaced by another. The overlap threshold $\theta = 0.1$ was fixed before the hold-out; the certification is insensitive to $\theta \in [0.05, 0.2]$ because a surviving entity’s frame-to-frame overlap greatly exceeds θ except at genuine loss.

Kill-switch and power. Before the deep-turnover certification we performed a development-only power analysis: from pilot variance components we asked what sample size could place a lower confidence bound above the threshold under alternatives $R^2 \in \{0.50, 0.65, 0.75\}$. When the pilot point estimate itself lies at or below the threshold, no finite sample can raise the lower bound above it; adequate power then means enough samples to show the effect is *below* threshold with confidence, which is the kill-switch’s designed function. The frozen prospective sample was fixed accordingly, and the family was executed once.

The experimental sequence. The results below are the outcome of nine preregistered experiments. (1) A material-continuity experiment established that the droplet persists under turnover but that a snapshot nearly suffices to predict its future, i.e. no strong history-specific hidden persistence. (2) A hidden-state experiment showed the internal state is a generic causal attractor, not an individuating fingerprint. (3) An organizational-memory experiment introduced the designed two-component field and showed it causal, erasable, transplantable, with a readout that collapsed onto one cumulative coordinate. (4) A multi-channel experiment added the $m_- \rightarrow$ attractant readout and made temporal order causally readable. (5) A writing-dimensionality audit corrected an apparent high-dimensional storage via grouped evaluation. (6) A writing redesign (C1c) produced two decodable coordinates at rest but a one-dimensional causal response. (7) A spatial-memory experiment characterized h_2 as a frame-free dispersion and first reported a turnover collapse. (8) A maintenance experiment corrected that report — the collapse was family-dependent and underpowered, the loss neither homogenization nor dilution — but did not certify deep survival. (9) A final kill-switch certified, on a sealed family, that h_2 does not survive deep turnover. The tracker-continuity audit then re-certified all turnover claims longitudinally.

5 Results

We present the ladder in the order it was climbed, so that both the corrections and the final numbers are visible.

Persistent organization; a generic hidden attractor. The droplet remains a localized, viable entity as its material turns over. A hidden internal state exists but behaves as a generic causal attractor rather than a history-specific identity (individuation area-under-curve ≈ 0.3 – 0.4 , i.e. below chance for history discrimination), consistent with the dynamical-systems expectation that recoverability of a hidden coordinate does not imply individuation. Two further controls guard

Table 3: Causal-intervention conditions.

condition	operation	tests
erase	set $M_f = 0$	memory necessity
mean transplant	copy entity-mean (m_1, m_2) into fresh body	transplantability (scalar)
structured transplant	copy entity-relative pattern into fresh body	spatial transplantability
$\lambda_+ \rightarrow 0$	disable the uptake readout	channel specificity (uptake)
$\lambda_- \rightarrow 0$	disable the attractant readout	channel specificity (attractant)
both $\rightarrow 0$	memory-inert control	causal necessity (all channels)
exact clone	re-run under bounded environmental noise	stochastic ceiling

Table 4: History families (each two-phase; amplitudes sampled independently in the stated band).

family	phase amplitude band	purpose	h_1, h_2
F_{mid}	[0.003, 0.020]	viable, mildly saturating	Eq. (5)
F_{low}	[0.001, 0.008]	more graded (capacity-viability map)	Eq. (5)
matched-dose (order)	phase-swapped $N \rightarrow c$ vs $c \rightarrow N$	isolate temporal order	discrete

against trivial explanations of turnover results. An *environmental-reset* control standardizes the nutrient and attractant fields at interrogation so a decode cannot ride residual gradients; h_1 survives it, following internal memory rather than environment. A *material-versus-organization* control decodes h_1 separately from memory on old and on newly grown constituents: both decode h_1 essentially equally at intermediate turnover, so new material acquires the memory distribution rather than merely diluting it — though, as the discussion notes, this acquisition is a passive consequence of $M_f \neq g m$, not active reconstruction [16, 18].

An engineered causal memory h_1 . The two-component field is causal and erasable: ablating the coupling ($\lambda = 0$) removes the effect exactly, and the memory transplants into a fresh standardized body and is turnover-related. Figure 2 shows the held-out h_1 causal response after mean transplantation into a fresh body ($R^2 = 0.61$), the in-place memory-isolated effect ($R^2 = 0.50$), and the memory-inert control, which collapses the response — the interventional signature of a causal memory [4]. The magnitude of the transplant effect (0.61) is smaller than the rest-state storage decode (0.94) precisely because the mean-field transplant discards spatial structure and reads only the scalar entity means; the point is not the magnitude but the interventional dissociation between the active and inert conditions, which no residual environment or body-size confound can produce (the recipient body is standardized and erased).

Temporal order made readable. A second orthogonal readout ($m_- \rightarrow$ attractant) makes matched-dose temporal order causally readable: the order contrast lives on the attractant axis, is essentially absent on the uptake axis, and collapses by a factor of ~ 70 – 73 under λ_- ablation (Figure 5). This establishes that the two readouts are channel-specific and that temporal order can be *written and read*.

Table 5: Sealed prospective families (generated and hashed before selection; executed once). Seed families are disjoint.

experiment	dev seeds	prospective seeds	SHA-256 (prefix)
Phase C (writing)	34001-3	35001-3	c6d0cd3c
SMC (spatial)	36001-3	36501-3	2d6986fe
DMM (maintenance)	37001-3	37501-3	923d8890
H2-CERT (kill-switch)	—	38501-4	4265b98c

Apparent dimensionality corrected by grouped evaluation. An initial continuous-history analysis appeared to recover a high-dimensional history. This was an artefact of two compounded errors: the two “independent” coordinates were driven with $20\text{--}47\times$ mismatched amplitudes, and a row-wise cross-validation leaked through replicates that shared a history [7]. Grouped leave-history-out evaluation reduces the headline continuous R^2 from 0.57 to 0.19, and three independent estimators (collinearity, a sensitivity singular-value ratio, and effective participation rank) agree that the frozen writing map is of rank ≈ 1 (Figure 6). This correction is a core methodological result: internal variation and a leaky decoder can jointly manufacture apparent high-dimensional storage where none exists.

Two-coordinate storage at rest, but a one-dimensional causal response. A minimal de-saturating change to the writing gains and timescales (the frozen “C1c” configuration; Table 2) yields two prospectively decodable coordinates *at rest* ($h_1, h_2 \approx 0.94$ held-out), with the second-dimension singular ratio rising from ≈ 0.02 to ≈ 0.23 . However, storage is not causal expression: with the existing readouts and a mean-field transplant, h_1 is causally expressed (0.61 transplant, 0.50 in-place) whereas h_2 is not (-0.04 transplant, 0.30 in-place), and the causal-response effective dimensionality remains ≈ 1 (Figure 6). The rank ratio at prospective evaluation (0.283) is marginally below the preregistered 0.30 gate, which we report as a strict, marginal failure rather than round up. The dissociation between storage and causal expression is the crux of the programme. The de-saturated writing genuinely stores two coordinates — a fast component that tracks the recent drive and a slow component that tracks the cumulative drive — so that at rest a decoder reading the memory field recovers both h_1 and h_2 . Yet the physics only “sees” the memory through two scalar readouts, and those readouts, at the mean-field level probed by transplantation, express only the cumulative combination. Storing a distinction that the dynamics cannot act upon is not functional memory in any operational sense; it is a latent variable with no downstream consequence. This is why causal expression is a separate, load-bearing gate rather than a corollary of a high rest-state decode.

The second coordinate is a frame-free dispersion; the deep-turnover kill-switch. h_2 is carried by a frame-free *value-dispersion* statistic of the memory (standard deviation and percentiles over entity cells), not by a resolvable geometric pattern: a laboratory-frame decoder outperforms it at rest but collapses under random translation, whereas the frame-free decoder is invariant by construction. The turnover behaviour of h_2 is highly family-dependent, and an adequately powered, preregistered kill-switch on a sealed family executed once returns a deep-turnover held-out $h_2 \approx 0$ at $M \approx 0.21$ (Figure 7). Following the kill-switch rule, the h_2 escalation was closed on this architecture. Before closing, we decomposed the loss to avoid mislabelling its cause. Contrary to an intuitive expectation, disabling the two spatial smoothing terms (templating and diffusion)

individually or together left the h_2 decode trajectory essentially unchanged, and the memory value–dispersion in fact *grows* during turnover rather than homogenizing; a dedicated unit test confirmed the ablations were physically active (they alter the one–step memory field) but negligible on the decode because growth injects competing dispersion. Nor is the loss dilution: old and new material decode h_2 equally at intermediate turnover, because new material inherits the local memory. A four–seed pooled analysis retained h_2 to $M \approx 0.58$ (CI [0.57, 0.97]), while a single earlier seed family had appeared to collapse — establishing high between–family variance and motivating the adequately powered, sealed kill–switch that finally returned $h_2 \approx 0$ at $M \approx 0.21$ (Table 6).

5.1 Tracker–continuity incident and correction

Human visual inspection of an animation of the frozen turnover trajectory (seed 38501) revealed the tracked–entity overlay “teleporting” between droplets, with a synchronized discontinuity in the h_2 time series and red contour at both horizontal edges. This prompted a dedicated audit, which we report in full because it changed the scientific record.

Diagnosis. The selection rule everywhere in the historical code was $\text{entity} = \arg \max_e \text{size}(e)$, evaluated independently at each checkpoint — a per–frame *reselection*, not longitudinal tracking. An event log on seed 38501 found five hard switches (centroid jumps of 19–32 cells with zero mask overlap), and both M and $\text{std}(m_-)$ jumped at exactly those frames; M even became non–monotonic. Connected–component detection, however, is periodic and the memory features are permutation–invariant: whole–world translations (including boundary–straddling shifts) leave entity size and all memory statistics bit–invariant, only the display centroid moving. The “red at both edges” is therefore a single periodically–wrapping entity, correctly detected, and the teleport is reselection, not a wrap artefact.

Correction and hold–out certification. We froze a longitudinal tracker (initialize once; match by maximum periodic mask overlap; censor on loss; never use h_1 , h_2 , M , or labels) and re–certified on *untouched* sealed seeds 38502–38504 (seed 38501, used to discover and develop the correction, was excluded as primary). On these hold–out seeds the tracker eliminates switching (0 switches, 36/36 track survival) and certifies h_1 : initial $R^2 = 0.92$ ([0.86, 0.99]), deep $R^2 = 0.98$ ([0.97, 1.00]) at $M \approx 0.19$, while h_2 remains 0.34 (the confidence interval is over 12 donor–group histories, not individual rows), (< 0.50) (Figure 4, Table 7). Critically, h_1 decodes $\approx$ equally from the largest, the second–largest, and the longitudinally–tracked entity (0.96/0.98/0.99), because h_1 is a *global* coordinate; the load–bearing result is therefore robust to tracker choice, and this robustness is emphatically *not* evidence of individual memory. A code–path audit confirmed that every single–timepoint and common–body causal/transplant claim (Figure 2, Figure 5) never traversed a turnover trajectory and is therefore unaffected by the reselection error (Table 7).

6 Discussion

The ladder is designed to separate concepts that are routinely conflated, and the value of the paper is as much in the separations as in the final numbers. *Memory versus decodability*: a coordinate that is decodable at rest (h_2) need not be a memory in any causal or persistent sense. *Storage versus causal expression*: with the frozen readouts, two decodable coordinates at rest coexisted with an $\approx$ one–dimensional causal response; storing a number the system cannot act on is not yet functional memory [4]. *Passive copying versus reconstruction*: because new material inherits the

Table 6: Primary results and 95% confidence intervals (prospective, grouped leave-history-out unless noted).

quantity	value	95% CI	status
h_1 at rest (sealed)	0.94	[0.89, 0.99]	established
h_1 deep turnover, longitudinal (hold-out)	0.98	[0.97, 1.00]	certified
h_1 causal (mean transplant / in-place)	0.61 / 0.50	—	established
h_1 vs body baseline at $M \approx 0.21$	0.93 vs 0.64	—	established
h_2 at rest (sealed)	0.90	[0.41, 0.97]	decodable
h_2 deep turnover, longitudinal	0.34	spans 0.5	fail (< 0.50)
h_2 causal (transplant / in-place)	-0.04 / 0.30	—	fail
order contrast λ_- ablation	$\sim 70-73\times$	—	established
longitudinal track survival to $M \leq 0.25$	36/36	—	established

Table 7: Tracker sensitivity and claim impact (deep turnover, untouched seeds 38502–38504).

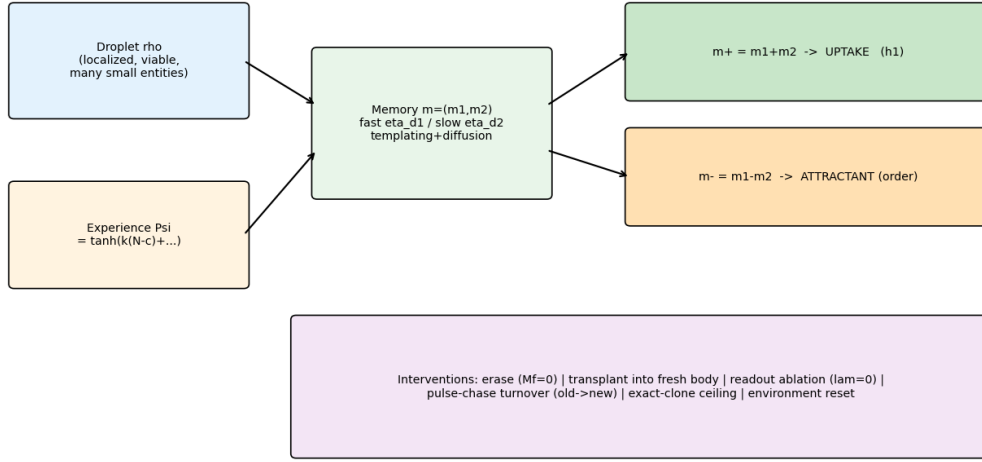
tracker / claim	h_1	h_2	impact
largest-at-each-frame (historical)	0.96	-0.18	reselection; teleports
second-largest blob	0.98	—	h_1 tracker-independent
longitudinal (corrected)	0.98	0.34	0 switches, 36/36 survive
transplant / ablation / endpoint causal	—	—	single-timepoint: unaffected
old-material M , size trajectory	—	—	replaced by longitudinal

local intensive memory during growth, propagation of a coordinate into new material is guaranteed by construction and is *not* evidence of active reconstruction toward a target. *Material replacement versus organizational inheritance*: h_1 information persists in the internal memory when body size no longer carries it, which is why it exceeds trivial body baselines after turnover. *Visual/size-reselected continuity versus longitudinal continuity*: a largest-component tracker manufactured apparent continuity that a longitudinal tracker removes [35, 36].

What does h_1 establish, and what does it not? It establishes that an engineered, bounded internal state can be a *causal*, *transplantable*, and *turnover-surviving* record of cumulative experience in a persistent dissipative droplet, certified prospectively on untouched seeds under continuity-safe tracking. It does *not* establish individual memory. Because the history is imposed globally, h_1 is written similarly into every large droplet, so it is decodable from any of them; its informational content is global even though its storage and causal expression are local. This is precisely why h_1 is robust to the choice of tracker — and why that robustness must not be misread as individuation. A genuine individuating memory would require different droplets in one world to carry *distinct* histories and be distinguishable by them, which our globally driven design does not test and does not exhibit.

Why did the second dimension fail? h_2 is carried by the value-dispersion of the memory, and the maintenance terms in Eq. (2) — local templating toward a neighbourhood mean and weak diffusion — are linear low-pass operators that erode dispersion, while ongoing growth injects fresh, history-independent heterogeneity that competes with the h_2 signal. Molecular memory that survives division uses *bistable*, hysteretic maintenance [25–28], not linear averaging; our architecture has no such protection, and an adequately powered kill-switch confirmed that h_2 does not survive deep turnover. This is a clean, informative negative: it delimits what the tested architecture can support and identifies the specific ingredient (protected, nonlinear maintenance) a future architecture would

Figure 1 — Droplet model, memory channels, and causal-intervention ladder



C1c frozen architecture; commit 1f8f789. Memory is ENGINEERED (not spontaneous, not identity).

Figure 1: **Model and causal-intervention ladder (schematic).** Question: what are the write/read pathways and interventions? Frozen C1c architecture: experience signal Ψ (Eq. 1); two-timescale memory $m = (m_1, m_2)$ (Eq. 2); readouts $m_+ \rightarrow \text{uptake}$ and $m_- \rightarrow \text{attractant}$ (Eqs. 3–4); and the intervention suite (erase, transplant, ablation, pulse-chase, clone). Schematic; no data. Source: docs/paper/fig1_model_ladder.png.

need.

The methodological contribution generalizes beyond this system. Each rung illustrates a failure mode that recurs in computational and empirical science: high internal variation mistaken for stored information; row-wise cross-validation leaking through grouped replicates [7]; a single favourable seed family taken as general; an underpowered estimate at the boundary of a threshold; and a size-based reselection mistaken for identity tracking. Preregistration, grouped evaluation, sealed hold-outs, kill-switches, and continuity-safe tracking are the countermeasures, and the paper is in part a worked example of applying them [32–34].

Finally, the results speak to the architecture-versus-substrate distinction. We do not claim that the Particle-Dynamics substrate is incapable of multidimensional organizational memory; we claim that the tested C1c writing/maintenance/readout architecture did not support a qualified second dimension, and that individuation was not demonstrated. The substrate-general question remains open, and the failed h_2 points to concrete, non-implemented directions — active local reconstruction, bistable/epigenetic-like maintenance, or compartmentalized maintenance — rather than to a claim of impossibility.

It is worth stating explicitly why the global nature of h_1 is not a defect of the experiment but the correct reading of what was measured. In a world driven by a single global history, the “memory of experience” that any droplet can carry is, at best, a local copy of that shared history; there is no per-droplet private experience to individuate on. Demonstrating individuation would require a fundamentally different design — distinct, locally imposed histories for co-existing droplets in one world, and a decoder that must attribute each droplet’s memory to its own history and not its neighbours’. Our design deliberately does not attempt this, and the tracker-sensitivity result (identical h_1 from different droplets) makes the reason vivid: when the signal is global, following the “right” droplet is unnecessary for decoding, which is exactly why it would be necessary — and currently absent — for any individuation claim. The honest conclusion is therefore narrow and, we

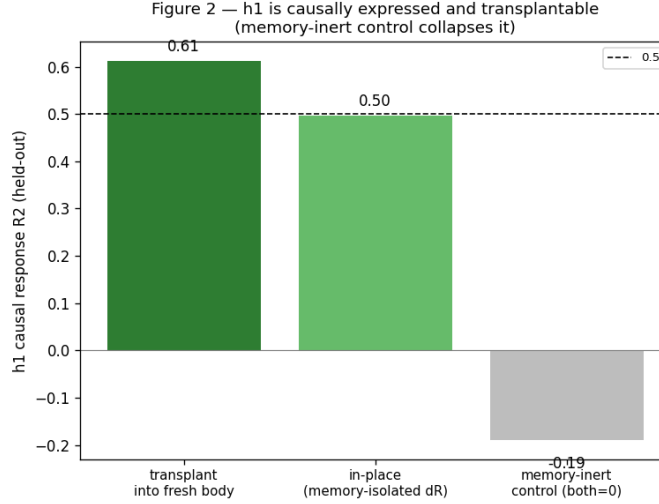


Figure 2: h_1 is **causally expressed and transplantable**. Protocol: mean transplant of donor memory into a fresh standardized erased body, in-place memory-isolated response (full minus memory-inert), and the memory-inert control. Prospective, grouped leave-history-out, $n = 12$ histories. Held-out h_1 response $R^2 = 0.61$ (transplant), 0.50 (in-place); the memory-inert control collapses it (causal necessity). Uncertainty: donor-level. Limitation: mean-field readout. Data: `results/wd01_phasec/phasec_causal_*.raw.pkl`; script `causal_runner.py`.

believe, defensible: an engineered internal state can be a causal, transplantable, turnover-surviving record of a *shared* history in a persistent droplet, and this is worth reporting precisely because it stops short of, and clearly delimits, the stronger claims it is often mistaken for. For the artificial-life programme [31], the practical lesson is a checklist for memory claims in simulated organisms: require causal intervention, not decodability; require turnover survival under continuity-safe tracking, not visual persistence; distinguish global from individual information; and preregister a kill-switch so a marginal signal cannot license indefinite mechanism accretion. A qualified second, individuating dimension in this family of models would need three things the present architecture lacks: a second, physically *independent* write signal (so two coordinates are not two filters of one scalar); a *nonlinear*, *bistable* maintenance rule (so a stored distinction is protected against linear averaging and material renewal) [25, 27]; and a design with *distinct per-droplet histories* (so individuation is even testable). None is claimed here. More broadly, we expect the qualitative pattern — a global, cumulative coordinate surviving turnover while an order/dispersion coordinate does not — to generalize to related reaction-transport memories with linear maintenance, and we expect the specific corrections (grouped evaluation, continuity-safe tracking, kill-switched escalation) to transfer directly to any simulated-organism study that decodes a history variable from an internal state. What will not transfer without new physics is a positive individuation result: that requires the three missing ingredients above and a design in which co-existing entities carry genuinely distinct histories.

7 Limitations

Several limitations bound the interpretation and are stated plainly. First, the memory is *engineered*, not emergent: we did not show that such a memory arises spontaneously, only that, once present, it is causal and turnover-surviving. Second, the dynamics are *simulator-specific*; the quantitative thresholds and the particular maintenance terms are properties of this model, and although the

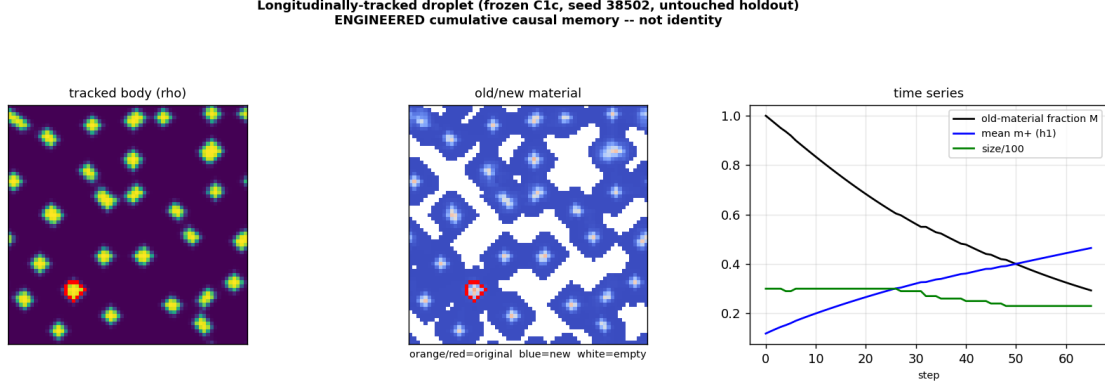


Figure 3: **Longitudinally-tracked turnover.** Question: does a single tracked droplet retain h_1 as its material is replaced? Protocol: frozen C1c, untouched seed 38502, longitudinal tracker. The tracked entity (red contour) is shown on the body panel and, with the *same* contour, on the old/new-material panel (orange/red = original, blue = new, white = empty); the time series shows old-material fraction M declining, mean m_+ (h_1) rising, and stable size. Single deterministic trajectory (illustrative); population certification is Figure 4. Data/script: `scripts/observe_droplet.py`.

Table 8: Development gate outcomes across the ladder (abbreviated).

gate	criterion	outcome
backward compatibility	$\lambda_- = 0$ bit-identical to predecessor	pass (0 deviation)
no tag/tracker leakage	no label in dynamics	pass
viability	localized entity through turnover	pass
material turnover	passive cohort M decreases	pass
storage rank (2nd dim)	$\sigma_2/\sigma_1 \geq 0.30$ (prospective)	marginal fail (0.283)
h_1 deep-turnover (longitudinal)	held-out $R^2 \geq 0.50$, lower CI > 0	pass (0.98, [0.97, 1.00])
h_2 deep-turnover	held-out $R^2 \geq 0.50$	fail (0.34)
individuation	distinct per-droplet histories decodable	not tested / not supported

model is in the reaction-diffusion / active-droplet family [8, 19], no claim of quantitative correspondence to a physical protocell is made. Third, the entities are *small* (~ 30 – 55 active cells), so per-entity samples are limited and decoders operate near their statistical floor at deep turnover; the certification survives this, but it is a constraint. Fourth, the histories are *designed and global*; h_1 is consequently low-dimensional and global, and the design cannot test individuation, which requires distinct per-droplet histories. Fifth, memory is *explicitly copied* into new material during growth, so “inheritance” into new material is by construction, not reconstruction. Sixth, the causal transplant readout is *mean-field*; a structured transplant that preserves spatial pattern was not shown to rescue h_2 , and structured causal expression of a second dimension remains untested. Seventh, the historical turnover analyses used *largest-component reselection*; we corrected this and re-certified h_1 longitudinally, but the correction narrows the original “the entity survived” framing to “a continuously tracked droplet retains a global coordinate.” Eighth, we performed *no reproduction, budding, or lineage* experiment, and make no heredity or genome claim. Ninth, the second-dimension rank ratio at prospective evaluation (0.283) is marginally below the preregistered gate (0.30); we report this as a strict, marginal miss. Tenth, the tracker, though frozen and label-free, was developed with knowledge of one incident trajectory; we mitigated this by certifying on

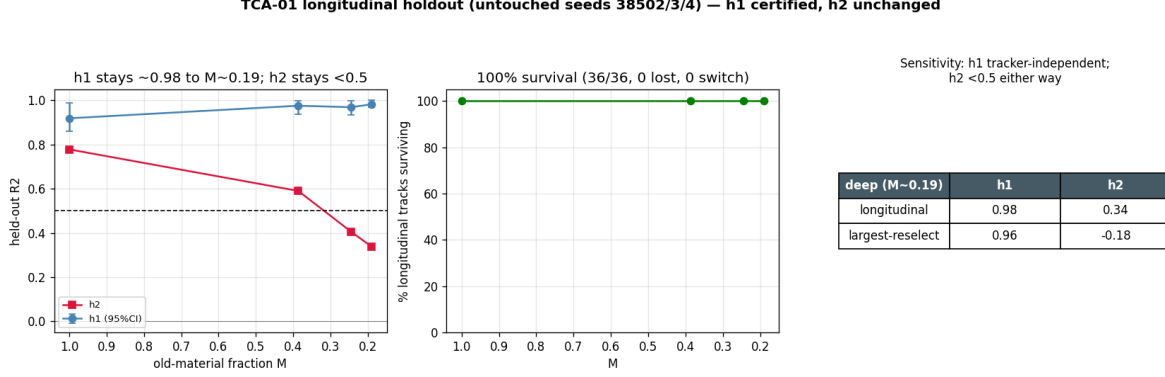


Figure 4: **Longitudinal hold-out certification.** Untouched sealed seeds 38502–38504, $n = 12$ histories $\times 3$ seeds, executed once. h_1 held-out R^2 with 95% CI versus M (stays ≈ 0.98 to $M \approx 0.19$); 100% track survival (36/36); sensitivity table (largest vs longitudinal). Prospective, grouped, donor-level bootstrap CIs. Result: h_1 certified; $h_2 < 0.5$. Data: `results/observer/tca_holdout_raw.pkl`.

Table 9: Post-incident claim corrections (tracker-continuity audit).

claim	status after correction
largest-component trajectories as unique continuity	invalidated
h_1 deep-turnover survival	replaced: longitudinal 0.98, [0.97, 1.00], 36/36
old-material M , entity-size time series	replaced by longitudinal
“48/48 viable”	qualified: 36/36 longitudinal survival
“a droplet carries memory”	qualified: a <i>continuously tracked</i> droplet
tracker-independent decoding	qualified: because h_1 is global, not individual
transplant / ablation / endpoint causal	unaffected (single-timepoint, common body)
h_2 turnover failure	unchanged ($0.34 < 0.50$ longitudinally)

disjoint, untouched seeds, but a fully independent re-seal would further strengthen the claim.

8 Conclusion

Under a globally imposed history, many persistent simulated droplets locally encode a similar engineered one-dimensional cumulative causal memory; a continuously tracked droplet retains this internal causal state through substantial constituent turnover (h_1 held-out $R^2 = 0.98$, [0.97, 1.00], at $M \approx 0.19$, 36/36 survival), and expresses it causally and transplantably. This memory does not individuate droplets: its informational content is global even though storage and causal expression are local. A second, initially decodable coordinate did not qualify as an independent, causal, turnover-surviving dimension, and was closed by a preregistered kill-switch. We report these boundaries — and the corrections required to reach them — as the result. Individuation, reproduction, and a genome are not warranted by the present evidence.

Author contributions

[AUTHOR NAME REQUIRED] designed and performed the simulations and analyses and wrote the manuscript. Author identity, affiliation, and ORCID are placeholders pending the corresponding

Figure 4 — Temporal order is stored in m_- and made causally readable (MCM-00)

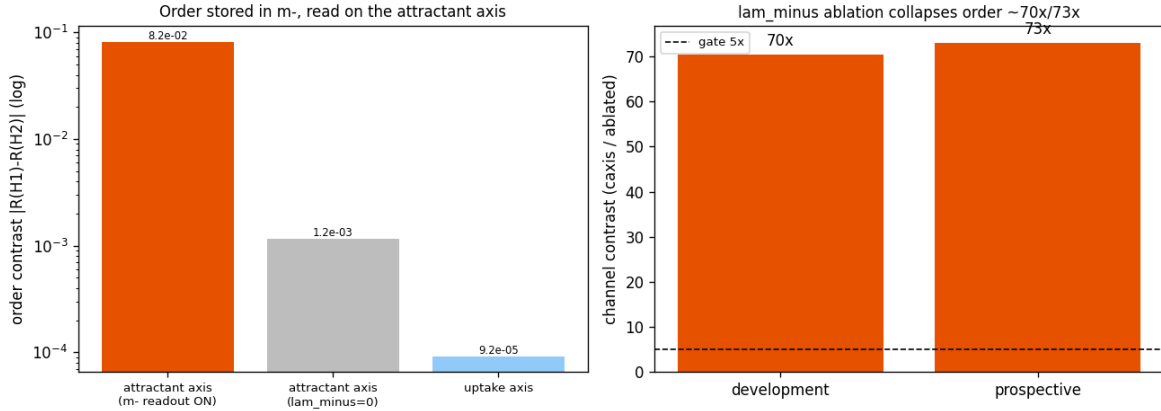


Figure 5: **Temporal order stored in m_- and made causally readable.** Order contrast on the attractant axis (readout on) vs λ_- -ablated vs the uptake axis, and the $\sim 70/73\times$ channel contrast (development / prospective, $n = 16/12$). Result: channel-specific order readout; ablation collapses it. Data: `results/sc_mcm/certificate.json`.

Table 10: Interpretation boundaries: what each result does and does not license.

observation	licenses	does not license
h_1 decodable + causal + transplantable	internal causal memory	individual identity
h_1 survives turnover (longitudinal)	organizational (not material) memory	lineage / heredity
h_1 global across droplets	memory of shared history	per-droplet private experience
h_2 decodable at rest	stored variation	causal or persistent memory
new material acquires h_1	passive local propagation	active reconstruction
kill-switch $h_2 \approx 0$ deep	architecture bound	substrate-general impossibility

author’s confirmation.

Competing interests

The author declares no competing financial or non-financial interests.

Funding

[FUNDING STATEMENT REQUIRED — no external funding is asserted.]

AI-assistance disclosure (draft)

An AI system (an autonomous coding/analysis agent) assisted with implementation, statistical analysis, figure generation, the tracker-continuity audit, and drafting under human direction. All scientific claims trace to committed code and data; the human author is responsible for the content. This disclosure is a draft for adaptation to the target venue’s policy.

WD-01 Phase C — minimal 2-D writing rule (C1c dynamic-range correction, same Ψ)

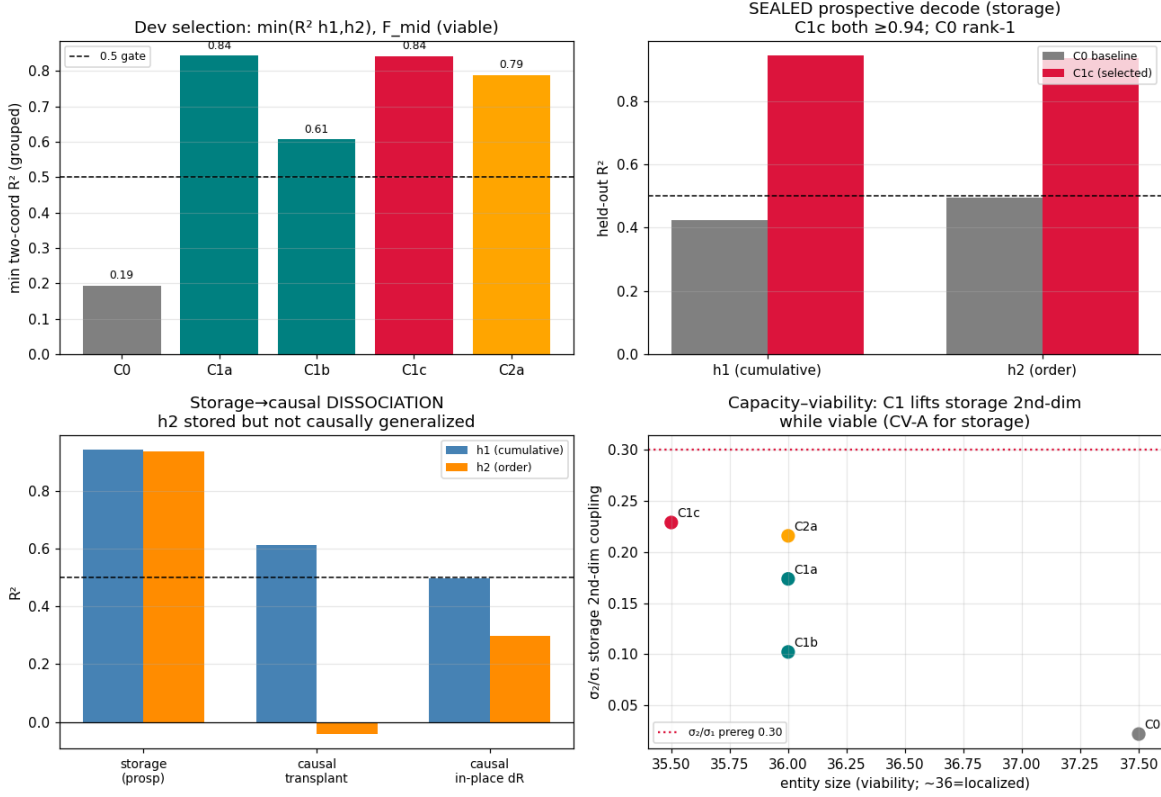


Figure 6: **Storage versus causal dimensionality.** Two prospectively decodable coordinates at rest with an $\approx$ one-dimensional causal response; the grouped-evaluation correction (row-wise 0.57 $\rightarrow$ grouped 0.19) and rank estimators. Prospective, grouped. Data: `results/wd01_phasesec/*_raw.pkl`.

Data and code availability

All code, frozen configurations, sealed prospective manifests (SHA-256), raw data, analysis scripts, and figure generators are retained under isolated version-control branches (Supplement). Negative results and corrections are preserved. No data were withheld.

Threats to validity and robustness

We enumerate the residual threats and how each is addressed. *Selection on the prospective family:* sealed families are hashed before selection and executed once; the deep-turnover certification uses seeds never inspected during tracker development. *Multiple comparisons:* the ladder is a fixed sequence of preregistered gates, not a search over many decoders; the single primary metric at each rung is declared in advance. *Decoder flexibility:* the primary decoder is a transparent regularized linear model, and nonlinear decoders are used only as secondary diagnostics, because a sufficiently flexible decoder can recover a target from nuisance structure and thereby manufacture a false positive. *Grouping:* all replicate rows sharing a history, a donor, a clone, or a trajectory prefix are held out together. *Tracker post-selection:* the longitudinal tracker uses only geometric overlap and is certified on disjoint untouched seeds. *Small samples:* the deep-turnover estimates are the

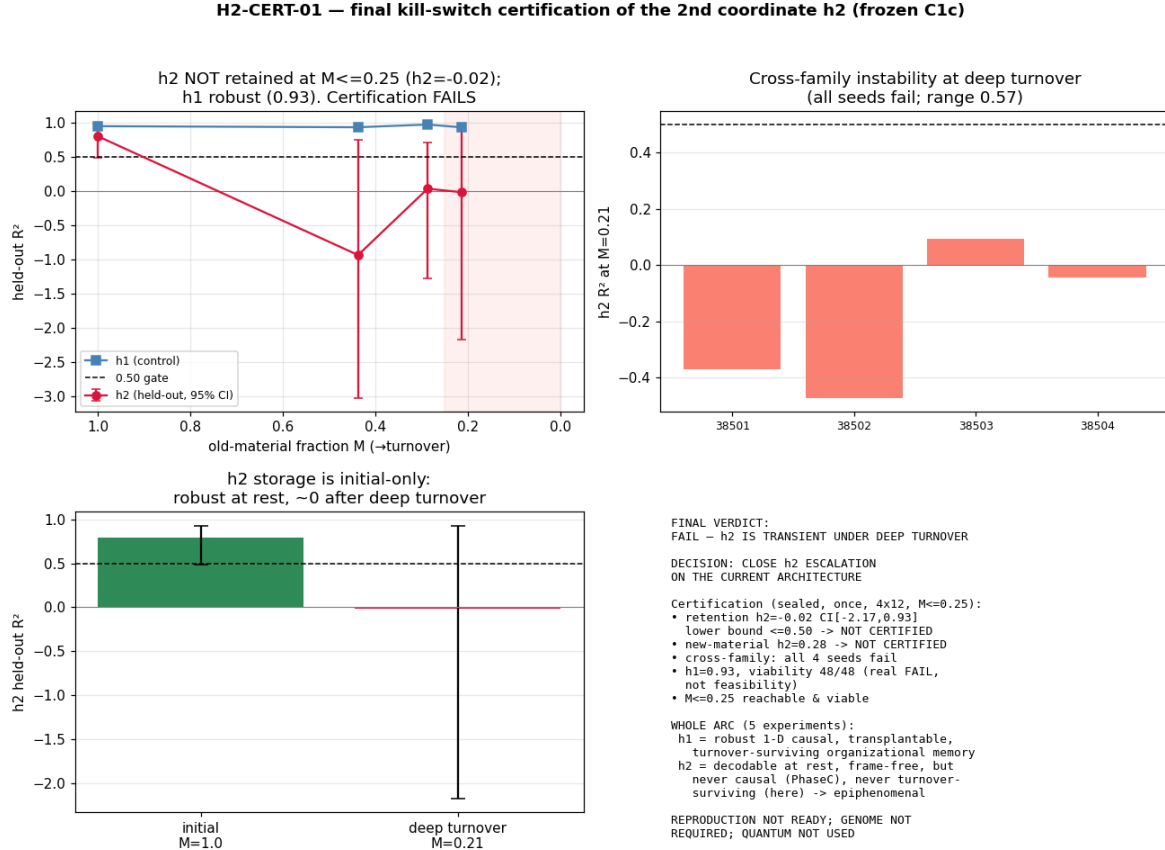


Figure 7: h_2 **deep-turnover kill-switch**. Sealed family executed once: held-out $h_2 \approx 0$ at $M \approx 0.21$ with 48/48 viability; family-dependence and moderate-turnover variability. Prospective. Data: `results/h2cert/h2cert_sealed_raw.pkl`.

weakest link, and we report their wide confidence intervals honestly rather than a point estimate; the h_1 certification survives this scrutiny because its lower bound is far from the threshold, whereas the h_2 failure is confirmed rather than manufactured by the small sample.

Reproducibility statement

Physics is frozen (Table 2); sealed families were hashed before selection and executed once (Table 5); decoding uses grouped leave-history-out cross-validation to avoid replicate leakage; the longitudinal tracker is frozen and label-free. Per-experiment run commands and hashes are in the Supplement.

References

- [1] W Ross Ashby. *Design for a Brain*. Chapman & Hall, 1952.
- [2] Humberto R Maturana and Francisco J Varela. *Autopoiesis and Cognition: The Realization of the Living*. D. Reidel, 1980.

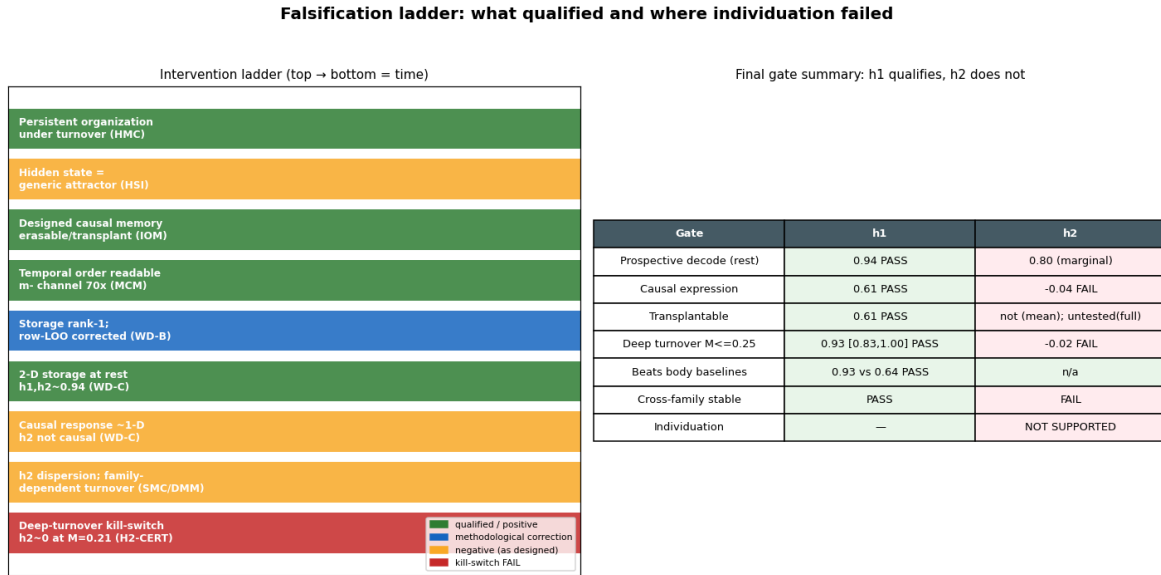


Figure 8: **Falsification ladder and final gate summary.** Intervention ladder (qualified / correction / negative / kill-switch) and the final gate table: h_1 qualifies on every gate, h_2 does not, individuation fails. Source: docs/paper/paper_claim_ladder.png.

- [3] Peter Godfrey-Smith. *Darwinian Populations and Natural Selection*. Oxford University Press, 2009.
- [4] Judea Pearl. *Causality: Models, Reasoning, and Inference*. Cambridge University Press, 2nd edition, 2009.
- [5] Donald B Rubin. Estimating causal effects of treatments in randomized and nonrandomized studies. *Journal of Educational Psychology*, 66(5):688–701, 1974. doi: 10.1037/h0037350.
- [6] Judea Pearl. Causal diagrams for empirical research. *Biometrika*, 82(4):669–688, 1995. doi: 10.1093/biomet/82.4.669.
- [7] Shachar Kaufman, Saharon Rosset, Claudia Perlich, and Ori Stitelman. Leakage in data mining: formulation, detection, and avoidance. *ACM Transactions on Knowledge Discovery from Data*, 6(4):1–21, 2012. doi: 10.1145/2382577.2382579.
- [8] Alan M Turing. The chemical basis of morphogenesis. *Philosophical Transactions of the Royal Society of London B*, 237(641):37–72, 1952. doi: 10.1098/rstb.1952.0012.
- [9] Alfred Gierer and Hans Meinhardt. A theory of biological pattern formation. *Kybernetik*, 12(1):30–39, 1972. doi: 10.1007/BF00289234.
- [10] Lewis Wolpert. Positional information and the spatial pattern of cellular differentiation. *Journal of Theoretical Biology*, 25(1):1–47, 1969. doi: 10.1016/S0022-5193(69)80016-0.
- [11] Mark C Cross and Pierre C Hohenberg. Pattern formation outside of equilibrium. *Reviews of Modern Physics*, 65(3):851–1112, 1993. doi: 10.1103/RevModPhys.65.851.
- [12] Shigeru Kondo and Takashi Miura. Reaction-diffusion model as a framework for understanding biological pattern formation. *Science*, 329(5999):1616–1620, 2010. doi: 10.1126/science.1179047.

- [13] Gregoire Nicolis and Ilya Prigogine. *Self-Organization in Nonequilibrium Systems*. Wiley, 1977.
- [14] Richard FitzHugh. Impulses and physiological states in theoretical models of nerve membrane. *Biophysical Journal*, 1(6):445–466, 1961. doi: 10.1016/S0006-3495(61)86902-6.
- [15] Jin Nagumo, Suguru Arimoto, and Shuji Yoshizawa. An active pulse transmission line simulating nerve axon. *Proceedings of the IRE*, 50(10):2061–2070, 1962. doi: 10.1109/JRPROC.1962.288235.
- [16] John J Hopfield. Neural networks and physical systems with emergent collective computational abilities. *PNAS*, 79(8):2554–2558, 1982. doi: 10.1073/pnas.79.8.2554.
- [17] Steven H Strogatz. *Nonlinear Dynamics and Chaos*. Westview Press, 1994.
- [18] Robert Hermann and Arthur J Krener. Nonlinear controllability and observability. *IEEE Transactions on Automatic Control*, 22(5):728–740, 1977. doi: 10.1109/TAC.1977.1101601.
- [19] David Zwicker, Rabea Seyboldt, Christoph A Weber, Anthony A Hyman, and Frank Jülicher. Growth and division of active droplets provides a model for protocells. *Nature Physics*, 13(4):408–413, 2017. doi: 10.1038/nphys3984.
- [20] Clifford P Brangwynne, Christian R Eckmann, David S Courson, Agata Rybarska, Carsten Hoege, Jobin Gharakhani, Frank Jülicher, and Anthony A Hyman. Germline p granules are liquid droplets that localize by controlled dissolution/condensation. *Science*, 324(5935):1729–1732, 2009. doi: 10.1126/science.1172046.
- [21] Anthony A Hyman, Christoph A Weber, and Frank Jülicher. Liquid-liquid phase separation in biology. *Annual Review of Cell and Developmental Biology*, 30:39–58, 2014. doi: 10.1146/annurev-cellbio-100913-013325.
- [22] Salman F Banani, Hyun O Lee, Anthony A Hyman, and Michael K Rosen. Biomolecular condensates: organizers of cellular biochemistry. *Nature Reviews Molecular Cell Biology*, 18(5):285–298, 2017. doi: 10.1038/nrm.2017.7.
- [23] Jack W Szostak, David P Bartel, and Pier Luigi Luisi. Synthesizing life. *Nature*, 409(6818):387–390, 2001. doi: 10.1038/35053176.
- [24] Martin M Hanczyc, Shelly M Fujikawa, and Jack W Szostak. Experimental models of primitive cellular compartments: encapsulation, growth, and division. *Science*, 302(5645):618–622, 2003. doi: 10.1126/science.1089904.
- [25] Timothy S Gardner, Charles R Cantor, and James J Collins. Construction of a genetic toggle switch in escherichia coli. *Nature*, 403(6767):339–342, 2000. doi: 10.1038/35002131.
- [26] Michael B Elowitz and Stanislas Leibler. A synthetic oscillatory network of transcriptional regulators. *Nature*, 403(6767):335–338, 2000. doi: 10.1038/35002125.
- [27] James E Ferrell. Self-perpetuating states in signal transduction: positive feedback, double-negative feedback and bistability. *Current Opinion in Cell Biology*, 14(2):140–148, 2002. doi: 10.1016/S0955-0674(02)00314-9.
- [28] Mark Ptashne. *A Genetic Switch: Phage Lambda Revisited*. Cold Spring Harbor Laboratory Press, 3rd edition, 2004.

- [29] Francisco G Varela, Humberto R Maturana, and Ricardo Uribe. Autopoiesis: the organization of living systems, its characterization and a model. *BioSystems*, 5(4):187–196, 1974. doi: 10.1016/0303-2647(74)90031-8.
- [30] W Ross Ashby. Principles of the self-organizing dynamic system. *Journal of General Psychology*, 37(2):125–128, 1947. doi: 10.1080/00221309.1947.9918144.
- [31] Mark A Bedau, John S McCaskill, Norman H Packard, Steen Rasmussen, et al. Open problems in artificial life. *Artificial Life*, 6(4):363–376, 2000. doi: 10.1162/106454600300103683.
- [32] Bradley Efron. Bootstrap methods: another look at the jackknife. *Annals of Statistics*, 7(1): 1–26, 1979. doi: 10.1214/aos/1176344552.
- [33] Brian A Nosek, Charles R Ebersole, Alexander C DeHaven, and David T Mellor. The preregistration revolution. *PNAS*, 115(11):2600–2606, 2018. doi: 10.1073/pnas.1708274114.
- [34] Karl R Popper. *The Logic of Scientific Discovery*. Hutchinson, 1959.
- [35] Azriel Rosenfeld and John L Pfaltz. Sequential operations in digital picture processing. *Journal of the ACM*, 13(4):471–494, 1966. doi: 10.1145/321356.321357.
- [36] Khuloud Jaqaman, Dinah Loerke, Marcel Mettlen, Hirotaka Kuwata, Sergio Grinstein, Sandra L Schmid, and Gaudenz Danuser. Robust single-particle tracking in live-cell time-lapse sequences. *Nature Methods*, 5(8):695–702, 2008. doi: 10.1038/nmeth.1237.